CSQ413 Seminar Abstract

Submitted by Arjun Manoj | Roll Number: 22 | CEC23CS043 | S7 CS - A

Topic 1 - Real-Time Detection Method for Cracked Eggs Based on Improved YOLOv8 and Dual Verification With Triple Classification Granularity

Base Paper Details

Authors: Daxu Sun, Hongfei Wei, Yangfan Luo, Weijian Li, and Yingshun Yu

Year: 2026

Journal: IEEE Access

Publisher: IEEE

DOI: 10.1109/ACCESS.2026.3686452

Abstract

Addressing the urgent need for real-time detection of cracked eggs on production lines, this study proposes a high-precision real-time detection algorithm based on an improved YOLOv8 framework. This study innovatively introduced a dual-detection method, incorporating a "crack region" classification to assist in distinguishing between intact eggs and cracked eggs. This method identifies cracked eggs by recognizing either local or overall crack characteristics, effectively reducing both false negatives and false positives. At the architectural level, the model innovatively integrates Weighted Bidirectional Feature Pyramid (BiFPN) to optimize multi-scale feature fusion, Deformable Convolutional Network v2 (DCNv2) to enhance perception of irregular geometric deformations in cracks, Weighted Intersection over Union v2 (WIoUv2) loss function to improve detection accuracy for occluded samples, and adds a small object detection layer to strengthen recognition of minute defects. Experiments demonstrate that the improved model achieves a mean average precision (mAP) of 92.9% on the test set, representing a 5.2% improvement over the original model and outperforming current mainstream detection models. In real-world production deployment testing, the system achieved a real-time detection speed of 45 FPS, with a false negative rate of only 0.46% and a false positive rate of 2.18% for intact eggs, while keeping the false positive rate for broken eggs below 2%. This performance far surpasses that of the original model and fully meets the requirements for industrial-grade real-time online quality inspection. This study provides an efficient solution for egg quality inspection with significant engineering application value.

Topic 2 - Automatic Psychophysiological Stress Detection Through Infrared Thermography and Explainable Artificial Intelligence

Base Paper Details

Authors: Irving A. Cruz-Albarran, Karla Muñoz Esquivel, Salvador Calderon-Uribe, Luis A. Morales-Hernandez, and Benjamin Dominguez-Trejo
Year: 2026
Journal: IEEE Access
Publisher: IEEE
DOI: 10.1109/ACCESS.2026.3676740

Abstract

Stress is the body's response to an unusual situation. While it can be beneficial to a certain extent, in large amounts, it can be adverse. This paper proposes a methodology for detecting psychophysiological stress using infrared thermography, deep learning and explainable artificial intelligence. The Trier Social Stress Test protocol was used to induce psychophysiological stress in 58 university students, and thermographic images were collected during this time. Subsequently, the facial and hand areas were automatically segmented. This information was used to train a convolutional neural network to detect psychophysiological stress. Similarly, the Grad-CAM method was implemented to identify which areas of the face or hands contributed to the classification. Finally, the computational complexity of the proposal was evaluated. The results demonstrate that stress detection can be achieved with 99.5% accuracy, 98.3% precision, 98.0% recall and 98.0% F1-Score. Regarding the Grad-CAM results, it was observed that both areas, the hand and facial areas, tend to contribute more to classification in women than in men. The computational complexity analysis of the model indicates that it used 480.371 MFLOPS, 11.569 MPARAMS and an average training time of 150.82 s. Based on this, it is concluded that infrared thermography is a technology that can aid in the detection of stress, and when combined with explainable artificial intelligence techniques, it enables users to understand the basis for their decisions, specifically which areas of the face and hands are affected.

Topic 3 - Homomorphic Encryption-Based Facial Recognition for Dual-Layer Similarity Analysis of Modified Images

Base Paper Details

Authors: Wooyoung Son, Soonhong Kwon, and Jong-Hyouk Lee

Year: 2025

Journal: IEEE Access

Publisher: IEEE

DOI: 10.1109/ACCESS.2025.3605615

Abstract

Recently, advanced closed-circuit televisions have been widely installed for security and crime prevention purposes, leading to the large-scale adoption of facial recognition systems. However, the indiscriminate use of such technologies has resulted in excessive intrusion into individuals' private lives, prompting ongoing discussions around privacy infringement. In response, Homomorphic Encryption (HE)-based facial recognition technologies which aim to ensure both privacy and security—have been actively researched and developed. Nevertheless, existing researches have limitations in that they did not seriously consider the possibility of original image leakage and failed to incorporate color information into the similarity measurement process. To address these issues, we propose a mechanism that considers not only facial landmarks but also color information in facial recognition, applying HE in a way that causes actual image modification in the facial area. This ensures that privacy can still be protected even in the event of facial data leakage. In the proposed mechanism, similarity values are independently derived from both facial landmarks and aligned color information. After decrypting each homomorphically encrypted value, a final Dual-Layer similarity is computed by combining both, enabling the identification of the facial image with the highest similarity. To achieve highly accurate and reliable final similarity computation, we propose a Dual-Layer similarity derivation formula based on the Softmax function, which is widely used as a core element in deep learning for multi-class classification. The effectiveness and necessity of the proposed method are demonstrated through comparisons with the arithmetic mean, weighted arithmetic mean, and harmonic mean. Furthermore, since most existing HE schemes are limited to supporting addition and multiplication operations, we analyze similarity derivation algorithms that comply with these constraints. Through a comparative analysis based on computational efficiency, we demonstrate that Euclidean Distance is well-suited for efficient HE-based facial recognition.